

# MAT1002: Calculus II L01

## Quiz 2 — 2025-2026, Term 2

Tuesday, 3 Mar 2026

1. (a) (6 points) Express the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1},$$

in the form of an elementary function. Identify its domain.

- (b) (10 points) Show that the Maclaurin series of  $f(x) := e^x$  converges to  $f(x)$  for all  $x \in \mathbb{R}$ .

**Solution:**

- (a) By Ratio Test,

$$\left| \frac{a_{n+1}}{a_n} \right| = \left( \frac{2n+1}{2n+3} \right) |x|^2 \rightarrow |x|^2,$$

as  $n \rightarrow \infty$ . The series converges when  $|x| < 1$  and diverges for  $|x| > 1$ . Let the function  $f$  be defined as  $f(x) := \sum (-1)^n x^{2n+1} / (2n+1)$  for  $-1 < x < 1$ . Consider its derivative

$$f'(x) = \sum_{n=0}^{\infty} (-1)x^{2n} = \sum_{n=0}^{\infty} (-x^2)^n = \frac{1}{1+x^2},$$

for  $|x| < 1$ . Then

$$f(x) = \int f'(x) dx = \int \frac{1}{1+x^2} dx = \arctan x + C,$$

where  $C$  is a constant. Using the fact that  $0 = f(0) = \arctan 0 + C = C$ , we know  $f(x) = \arctan x, x \in (-1, 1)$ . Note that it is okay to include  $x = 1$  (No penalty if students do not include endpoints).

- (b) The function  $f(x) = e^x$  has derivatives of all orders for  $x \in \mathbb{R}$ . Note that

$$f^{(k)}(x) = e^x \implies f^{(k)}(0) = 1, k \in \mathbb{N}.$$

By Taylor's theorem, we write

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^k}{k!} + R_k(x),$$

where the remainder

$$R_k(x) = \frac{e^c}{(k+1)!} x^{k+1},$$

for some  $c$  between 0 and  $x$ . Since  $e^x$  is an increasing function, so  $e^c$  lies between 1 and  $e^x$ . If  $x \leq 0$ , then  $e^x < e^c < 1$  and

$$|R_k(x)| \leq \frac{|x|^{k+1}}{(k+1)!}.$$

If  $x > 0$ , then  $1 < e^c < e^x$  and

$$|R_k(x)| \leq e^x \frac{|x|^{k+1}}{(k+1)!}.$$

Finally, because  $x^{k+1}/(k+1)! \rightarrow 0$  as  $k \rightarrow \infty$  for every  $x$ ,

$$\lim_{k \rightarrow \infty} R_k(x) = 0,$$

and the series converges to  $e^x$  for every  $x$ . Thus,

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

2. (4 points) Find the limit

$$\lim_{x \rightarrow 0} \left( \frac{1}{\sin x} - \frac{1}{x} \right).$$

**Solution:** We have

$$\begin{aligned} \lim_{x \rightarrow 0} \left( \frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \frac{x - \sin x}{x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{x - (x + \mathcal{O}(x^3))}{x(x + \mathcal{O}(x^3))} \\ &= \lim_{x \rightarrow 0} \frac{\mathcal{O}(x^3)}{x^2 + \mathcal{O}(x^4)} \\ &= \lim_{x \rightarrow 0} \frac{\mathcal{O}(x)}{1 + \mathcal{O}(x^2)} \\ &= \frac{0}{1 + 0} \\ &= 0. \end{aligned}$$

3. (a) (5 points) Calculate area of the surface generated by revolving about the  $x$ -axis the curve given by  $x = \cos t, y = 1 + \sin t, t \in [0, 2\pi]$ .
- (b) (5 points) Find the area under one arch of the cycloid  $x = r(\theta - \sin \theta), y = r(1 - \cos \theta), \theta \in [0, 2\pi]$ .

**Solution:**

- (a) For  $0 \leq t \leq 2\pi$ , the area is

$$S = \int_0^{2\pi} 2\pi y(t) \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2} dt = 2\pi \int_0^{2\pi} (1 + \sin t) \cdot 1 dt = (2\pi)^2.$$

- (b) The area is

$$\begin{aligned} A &= \int_0^{2\pi} y(\theta) \frac{dx(\theta)}{d\theta} d\theta \\ &= \int_0^{2\pi} r(1 - \cos \theta) r(1 - \cos \theta) d\theta \\ &= r^2 \int_0^{2\pi} (1 - 2\cos \theta + \cos^2 \theta) d\theta \\ &= r^2 \left( 2\pi + \frac{\cos 2\theta + 1}{2} \Big|_{\theta=0}^{2\pi} \right) \\ &= 3\pi r^2. \end{aligned}$$